High school · number theory

Congruences and last digits

Lesson and method

Before you begin

Divisibility, powers and Euclidean division.

Ideas and method

$a \equiv b$ (mod m), with $m \geq 2$, means that m divides $a - b$. They have the same remainder upon division by m. For example, $-1 \equiv 6$ (mod 7).

Congruences are preserved by addition and multiplication, hence by positive integer powers. Work modulo 10 to find a last digit.

A factor cannot always be cancelled: $2 \times 1 \equiv 2 \times 4$ (mod 6), but 1 is not congruent to 4 modulo 6. A factor coprime to the modulus can be cancelled.

Worked example

The last digits of $3^1, 3^2, 3^3, 3^4$ are 3,9,7,1. The cycle repeats because $3^4 \equiv 1$ (mod 10). Since $2026 = 4 \times 506 + 2$, the last digit of 3^{2026} is 9.

Explore

Build tables of squares modulo 3, 4 and 5. Identify impossible remainders.